

## 0.1 Power Series, Laurent Series, and the Residue Theorem

While Cauchy's Integral Formula provides an exact representation of a holomorphic function within a simply connected domain, many profound applications of complex analysis involve functions with isolated non-analytic points. To analyze these singularities, we extend the local Taylor series representation to the Laurent series, which accommodates negative powers of the complex variable.

[Laurent's Theorem] Let  $f$  be holomorphic in the open annulus  $A = \{z \in \mathbb{C} \mid r < |z - z_0| < R\}$ , where  $0 \leq r < R \leq \infty$ . Then  $f(z)$  can be uniquely represented by a uniformly convergent series on any closed sub-annulus of  $A$ :

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n$$

where the coefficients are given by:

$$c_n = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{(w - z_0)^{n+1}} dw$$

for any simple closed contour  $\gamma$  lying in  $A$  and enclosing the inner boundary  $r$ , oriented counterclockwise.

The series naturally decomposes into two parts: the analytic part  $\sum_{n=0}^{\infty} c_n (z - z_0)^n$ , which converges for  $|z - z_0| < R$ , and the principal part  $\sum_{n=1}^{\infty} c_{-n} (z - z_0)^{-n}$ , which converges for  $|z - z_0| > r$ . The behavior of the principal part provides a rigorous classification of isolated singularities.

Let  $f$  have an isolated singularity at  $z_0$ , and let  $\sum_{n=-\infty}^{\infty} c_n (z - z_0)^n$  be its Laurent expansion in the punctured disk  $0 < |z - z_0| < R$ .

1. If  $c_n = 0$  for all  $n < 0$ , the singularity is **removable**. The function can be extended holomorphically to  $z_0$  by setting  $f(z_0) = c_0$ .
2. If  $c_{-m} \neq 0$  for some integer  $m > 0$  and  $c_n = 0$  for all  $n < -m$ , the singularity is a **pole of order  $m$** .
3. If  $c_n \neq 0$  for infinitely many negative values of  $n$ , the singularity is an **essential singularity**.

The coefficient  $c_{-1}$  is of paramount importance. Termed the **residue** of  $f$  at  $z_0$  and denoted  $\text{Res}(f, z_0)$ , it is the sole term of the Laurent series that yields a non-zero integral when integrated over a closed loop enclosing  $z_0$ , because  $\oint_{\gamma} (z - z_0)^n dz = 0$  for all integers  $n \neq -1$ , and equals  $2\pi i$  for  $n = -1$ .

[Cauchy's Residue Theorem] Let  $\Omega \subset \mathbb{C}$  be a simply connected open set. Let  $\gamma$  be a simple closed contour in  $\Omega$ , oriented counterclockwise, and let  $f$  be holomorphic on  $\Omega$  except for a finite number of isolated singularities  $z_1, z_2, \dots, z_k$  located strictly in the interior of  $\gamma$ . Then:

$$\oint_{\gamma} f(z) dz = 2\pi i \sum_{j=1}^k \text{Res}(f, z_j)$$

By the principle of contour deformation (generalized Cauchy Integral Theorem for multiply connected domains), we construct small, non-overlapping circles  $\gamma_j$  centered at each singularity  $z_j$ , with each  $\gamma_j$  oriented counterclockwise and contained entirely within the interior of  $\gamma$ . The region bounded by  $\gamma$  and the reversed circles  $-\gamma_j$  contains no singularities of  $f$ . Thus, by Cauchy's Integral Theorem:

$$\oint_{\gamma} f(z) dz + \sum_{j=1}^k \oint_{-\gamma_j} f(z) dz = 0 \quad \oint_{\gamma} f(z) dz = \sum_{j=1}^k \oint_{\gamma_j} f(z) dz$$

For each  $j$ , the integral over the local circle  $\gamma_j$  isolates the residue at  $z_j$ . By expanding  $f(z)$  into its Laurent series around  $z_j$ , we find:

$$\oint_{\gamma_j} f(z) dz = \oint_{\gamma_j} \sum_{n=-\infty}^{\infty} c_{n,j} (z - z_j)^n dz = c_{-1,j} (2\pi i) = 2\pi i \operatorname{Res}(f, z_j)$$

Substituting this into the contour sum completes the proof.

The Residue Theorem provides a powerful analytic mechanism for evaluating purely real integrals, particularly improper integrals over the entire real line, bypassing the need for explicit antiderivatives.

**Application: Evaluating a Real Improper Integral** Evaluate the real integral  $I = \int_{-\infty}^{\infty} \frac{1}{x^2 + a^2} dx$  for  $a > 0$ .

We complexify the integrand, defining  $f(z) = \frac{1}{z^2 + a^2}$ . The function  $f$  has isolated singularities where  $z^2 + a^2 = 0$ , which factors as  $(z - ia)(z + ia) = 0$ . These are simple poles at  $z = ia$  and  $z = -ia$ . We define a closed semicircular contour  $\Gamma = [-R, R] \cup C_R$ , where  $[-R, R]$  is the segment on the real axis and  $C_R$  is the upper semicircle of radius  $R > a$  centered at the origin, traversed counterclockwise.

The only singularity of  $f$  enclosed by  $\Gamma$  is the simple pole at  $z = ia$  in the upper half-plane. We compute the residue at  $z = ia$ :

$$\operatorname{Res}(f, ia) = \lim_{z \rightarrow ia} (z - ia) \frac{1}{(z - ia)(z + ia)} = \lim_{z \rightarrow ia} \frac{1}{z + ia} = \frac{1}{2ia}$$

By the Residue Theorem, the contour integral evaluates to:

$$\oint_{\Gamma} f(z) dz = 2\pi i \operatorname{Res}(f, ia) = 2\pi i \left( \frac{1}{2ia} \right) = \frac{\pi}{a}$$

We decompose the contour integral into its linear and arc components:

$$\oint_{\Gamma} f(z) dz = \int_{-R}^R \frac{1}{x^2 + a^2} dx + \int_{C_R} \frac{1}{z^2 + a^2} dz$$

We bound the integral over the arc  $C_R$  using the estimation lemma (ML-inequality). For  $z \in C_R$ ,  $|z| = R$ . By the reverse triangle inequality,  $|z^2 + a^2| \geq |z|^2 - a^2 = R^2 - a^2$ . The length of  $C_R$  is  $\pi R$ . Thus:

$$\left| \int_{C_R} \frac{1}{z^2 + a^2} dz \right| \leq \frac{1}{R^2 - a^2} (\pi R)$$

As  $R \rightarrow \infty$ , this bound tends strictly to 0. Taking the limit as  $R \rightarrow \infty$  of the decomposed contour integral yields:

$$\frac{\pi}{a} = \lim_{R \rightarrow \infty} \int_{-R}^R \frac{1}{x^2 + a^2} dx + 0 = \int_{-\infty}^{\infty} \frac{1}{x^2 + a^2} dx$$

Thus, the improper integral converges to  $\frac{\pi}{a}$ , demonstrating the indispensable utility of analytic continuation and contour deformation in evaluating real-variable phenomena.